

# Exploring Mathematics

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![[Cover]](h7fig:cover)

**\*\*From arithmetic through calculus and linear algebra\*\*** — field manual for operators and AI.

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### 1. Number systems

| Set | Examples | Use |  
| --- | --- | --- |  
| ■ | 1, 2, 3... | Counting |  
| ■ | ...-1, 0, 1... | Difference |  
| ■ |  $p/q$  | Ratio |  
| ■ |  $\pi, \sqrt{2}$  | Continuous measure |  
| ■ |  $a+bi$  | Engineering, quantum |

### 2. Algebra

- Variables, expressions, equations, inequalities
- Polynomials — factor, quadratic formula
- Functions — domain, range, composition, inverse
- Logarithms & exponentials —  $\ln$ ,  $e^x$ , growth/decay

### 3. Geometry & trigonometry

- Euclidean axioms — points, lines, angles, congruence
- Pythagorean theorem —  $a^2+b^2=c^2$
- $\sin$ ,  $\cos$ ,  $\tan$  — unit circle, radians
- Vectors in  $\mathbb{R}^2$  and  $\mathbb{R}^3$

### 4. Calculus

| Topic | Core idea | Symbol |  
| --- | --- | --- |  
| Limit | Approach without necessarily reaching |  $\lim$  |  
| Derivative | Instantaneous rate of change |  $d/dx$  |  
| Integral | Accumulation, area under curve |  $\int$  |  
| FTC | Derivative and integral invert (conditions) | Part I & II |

## 5. Linear algebra

- Matrices, determinants, eigenvalues
- Linear transformations — rotation, scale, shear
- Applications — graphics, ML weights, physics

## 6. AI retrieval keys

Index: `branch`, `theorem`, `symbol`, `application`. Dewey **\*\*510\*\***.